

Rupjyoti Barman

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Summary

.NET Backend Developer with **3 years** of experience designing and delivering **hardware diagnostic APIs** and **system-level platforms** for **HP PC Hardware Diagnostics**, HP's official diagnostic application. Proven expertise in **C#, ASP.NET Core**, and **managed-unmanaged interoperability** (P/Invoke, Native C++ DLL integration), with hands-on experience in **decoupled NuGet-based architectures**, **WMI-based real-time monitoring**, and **Windows System APIs** (WUA, HPCMSL, DeviceIoControl). Experienced in **Clean Architecture**, **Factory Pattern**, **Entity Framework Core**, and **Azure Pipelines**, with a track record of eliminating third-party release dependencies, reducing diagnostic development cycles, and collaborating with **cross-functional global teams** across India, Taiwan, and the US.

Skills

- **Languages:** C#, C++, SQL, JavaScript, Python
- **Frameworks & Libraries:** ASP.NET Core (Web API, MVC), Entity Framework Core, WPF, React, Bootstrap, jQuery, LINQ
- **Windows APIs & Interop:** WMI, WQL, ManagementEventWatcher, WUA API, HPCMSL, DeviceIoControl, P/Invoke, Win32 APIs
- **Architecture & Design:** Clean Architecture, Layered Architecture, SOLID Principles, OOP, Factory Pattern, Dependency Injection, Controller-Service-Repository Pattern
- **Database, DevOps & Tools:** SQL Server, PostgreSQL, EF Core (Code-First, Migrations), CI/CD, Azure Pipelines, Git, NuGet, Visual Studio, Postman, Jira, Confluence, Serilog
- **Testing & Security:** MSTest, Moq, JWT Authentication, RBAC

Experience

Software Developer, BoldTek India Pvt. Ltd. - Guwahati

June 2023 - Present

Client: Hewlett-Packard (HP)

- Led the India team in migrating from a **shared RPC-based Windows Service architecture** to a **decoupled 3-layer NuGet package (Business Logic, C# Interop Wrapper via P/Invoke, Native C++ DLL)** with **Factory Pattern**-based dependency injection, eliminating shared service dependency across multiple HP teams; achieved **100% elimination of third-party release delays** (previously 10-15 days) and reduced **diagnostic development cycle from 4 to 3 sprints** (40 to 30 working days, a **25% reduction**) by removing 2 redundant architectural layers (Service layer and Service Manager routing layer), resolving cross-team unhandled exceptions and **sleep/hibernate-induced service failures** consumed by **HP PC Hardware Diagnostics**.
- Migrated **6 hardware diagnostic modules** (Storage: **DRV Read/Verify**, **SMART Health Monitor**, **SMART Data Collection**; Network: **NIC IRQ Verification**, **ROM/MAC Validation**; Audio: **Audio Playback Test**) from the legacy **RPC-based Windows Service** onto the new **NuGet architecture**; additionally engineered **2 diagnostic modules from scratch - Driver Update Detection (WUA API)** and **BIOS Update Check (HPCMSL, commercial systems)** - consolidating all into a **centralized diagnostic backend** architected to support multiple HP client applications with zero external service dependency.
- Replaced a service-based device discovery architecture (which required a **2-3 minutes of manual service restart** per hardware change) by engineering a **real-time WMI monitoring module (ManagementEventWatcher + WQL on Win32_PnpEntity)** packaged into the shared **Common** library; reduced device change detection latency **from 2-3 minutes to under 4 seconds (97% reduction)**, with the module adopted across multiple hardware diagnostic APIs.
- Configured **Azure Pipelines** to trigger **NuGet package builds**, publishing the **Business Logic** package (with required dependency NuGets) integrated into the **C# Wrapper** layer to produce the **final consolidated NuGet package** (bundling all layers including Native C++ DLL) for testing and client consumption.
- Verified all migrated and newly developed diagnostic modules using a dedicated **WPF-based API testing application** for end-to-end UI-level verification and **MSTest** unit test projects across **Business Logic** and **C# Wrapper** layers, leveraging **Moq** for dependency mocking to ensure diagnostic reliability prior to client integration.
- Engineered 2 production features in an **ASP.NET Core MVC** application (Controller, Service, Repository architecture with **EF Core Code-First** migrations and **SQL Server**) using **JavaScript** and **Bootstrap**; built an **ODM self-service dashboard** with 3-tier RBAC (Viewer, ODM, Admin) for INI configuration leveraging, verification submission, and approval status tracking, backed by **INI key-value validation logic** across **1000+ HP NPI configuration files**; developed a **Release Notes Dashboard** replacing manual Excel-based tracking, enabling the HP Factory Diagnostic team to query release-specific INI changes directly from **SQL Server**.
- Partnered with cross-functional global teams (**India, Taiwan, US**) to deliver features, debug production issues, and align on diagnostic API design across release cycles.

Education

B.Tech in Computer Science and Engineering - CGPA: 8.01/10

May 2023

North Eastern Regional Institute of Science and Technology (NERIST), Arunachal Pradesh